| PGDM 2025 - 2027 | | SPJIMR- Course outline | |
|---|---|---|---|
| **Course Name** | **Applied Business Analytics** | **Credits** | **2** |
| **Course Code** | **ANA545-PDM** | **Term** | **IV** |
| **Instructor(s)** | Dr. Aishvarya | | |
| **Lead Instructor** | Dr. Aishvarya | **TA/RA** | Medha Raikar |
| **No. of Sessions** | 18 | **No. of Group Work sessions** | |

**Course Description:**

This course provides an introduction to the field of business analytics, which includes working on business problems across domains and identifying the right analytical approach to address the problem. Students will learn numerous statistical and machine learning algorithms and work extensively with datasets and tools, programming languages enabling them to draw data-driven insights from the analysis.
Students will understand how managers use business analytics to formulate and solve business challenges, support managerial decision making, and apply the right statistical and/or machine learning algorithms. This course will benefit students who plan careers in analytics, targeted marketing, predictive modelling, and strategic consulting.

**Course Learning Outcomes:**

1. Understand and apply key analytical frameworks and tools to identify, analyse and solve real-world business problems across various industries
2. Gain hands-on experience with data wrangling, mining, and advanced analytics, enabling students to extract actionable insights from structured and unstructured data
3. Implement data-driven decision making and understand the challenges involved

**Prescribed Text:**

U. Dinesh Kumar, "Business Analytics – The Science of Data Driven Decision Making", Second Edition, Wiley, 2021

**Additional Readings:**

**Pedagogy:**

Lectures and Case Based Analysis

**Course Evaluation:**

| S. No. | Evaluation Component (Individual/ Group) | Evaluation Code | Weightage | Timeline for announcing evaluation | Scheduled Date of Evaluation | CLOs Mapped to Evaluations |
|---|---|---|---|---|---|---|
| 1 | Quiz – 1 | ANA545-PDM-46-I01 | 15% | Quiz-1 After Session 8 | | All |
| 2 | Quiz – 2 (Surprise Quiz) | ANA545-PDM-46-I02 | 15% | | | All |
| 3 | Group Assignment | ANA545-PDM-46-G01 | 40% | Roll out session 2-5 | | All |
| 4 | End Term Exam | ANA545-PDM-46-I03 | 30% | End of course | | All |
| | **Total** | | **100%** | | | |

Note: A surprise quiz can be conducted by the faculty at any point of time during the course. The weightage of the evaluation components will be adjusted accordingly

**Student Honour Code**

1. All students are expected to read the assigned readings before class. The course handout will include lessons, cases prescribed for sessions, and the class schedule, in addition to the prescribed readings.
2. Every student is expected to follow the ‘honesty’ policy in academic affairs and do their academic work. Any form of cheating or plagiarism constitutes unacceptable academic dishonesty. Please refer to SPJIMR's plagiarism policy.
3. Appropriate classroom behaviour is required for effective learning. These behaviours include timely arrival, attentiveness during class, and only authorized use of devices (per the faculty members’ prerogative).
4. SPJIMR is committed to continuously advancing its academic excellence. It is, therefore, mandatory for the student to provide feedback whenever the program requests it. Please note that all student feedback is anonymous.

**Session Details:**

| Session No & Faculty | Session Plan | Pedagogy | Suggested/Prescribed Readings | Mandatory Session |
|---|---|---|---|---|
| 1<br>Prof. Aishvarya | Introduction to Business Analytics and Data Mining<br>• Overview of descriptive, prescriptive and predictive analytics across multiple domains<br>• Understand Analytics Landscape: How different organizations have used analytics to solve business problems and compete in the market<br>• Overview of Supervised, Unsupervised and Reinforcement Learning<br>• Machine Learning and Statistical learning algorithms | Class Discussion | | |
| 2-3<br>Prof. Aishvarya | Predictive Analytics<br>• Simple and Multiple Linear Regression<br>• Data preparation; Training, Testing, Validation Dataset<br>• Model Diagnostic<br>• Model Evaluation and Improvement<br>**Group Assignment Roll out** | Case discussion<br>Hands on exercises using Python | Case: Pricing of players in the Indian Premier League (HBSP) | |
| 4-6<br>Prof. Aishvarya | Classification Algorithms and their Application:<br>• Binary Logistic Regression,<br>• Feature Selection, Diagnostic<br>• Multinomial Logistic Regression<br>**QUIZ – 1: Post Session 8** | Case discussion<br>Hands on exercises using Python | Case: HR Analytics at ScaleneWorks: Behavioral Modeling to Predict Renege | |
| 7-9<br>Prof. Aishvarya | Classification Algorithms and their Application:<br>• Optimal Cut-off Probability<br>• Classification and Regression Tree (CART) | Case discussion<br>Hands on exercises using R/ Python | Case: Predicting Earnings Manipulation by Indian Firms using Machine Learning Algorithms | |

| | | | | |
|---|---|---|---|---|
| | • Chi-Squared Automatic Interaction Detection (CHAID)<br>• Entropy and Gini Impurity Index<br>• K Nearest Neighbors (KNN) | | | |
| 10-12<br>Prof. Aishvarya | Clustering Algorithm and Use Cases:<br>• Distance and Similarity measures<br>• Market Basket Analysis<br><br>Introduction to Recommender Systems | Case discussion<br><br>Hands on exercises using R/ Python | Case: Customer Analytics at Big Basket-Product Recommendations | |
| 13-15<br>Prof. Aishvarya | Customer Analytics<br>• Estimating Customer Lifetime Value (CLV), Churn Probability and customer retention<br>• Understand Impact of Promotion on Brand Switching<br>• Markov Process and Reinforcement Learning<br><br>**Group Assignment Submission** | Case discussion<br><br>Hands on exercises using R/ Python | Case: Consumer Choice between House Brands and National Brands in Detergent Purchases at Reliance Retail | |
| 16-17<br>Prof. Aishvarya | Text Analytics for Business Decisions<br>• Text Pre-processing<br>• Sentiment Analysis<br>• Topic Modelling | Case Discussion<br><br>Hands on exercises using Orange | Case: Enhancing Visitor Experience at Iskcon using Text Analytics | |
| 17-18<br>Prof. Aishvarya | Integrating LLMs in Organizational Capability<br>• Use cases<br>• Hands-on exercise<br>• Using AI ethically and responsibly | Hands on exercises<br><br>Class Discussion | 13 Principles for Using AI Responsibly<br><br>Harvard Business Review (2023) | |

**AOL Alignment of the Course:**

**Alignment of intended Program Competency Goals & Course Learning outcomes:**

| Competency Goals (CGs) | Programme Learning Objectives (PLOs) *(with Bloom's Taxonomy)* | TLAs code (CLO) | Assessment code | Course learning outcomes | Asmt. method & weightage |
|---|---|---|---|---|---|
| **CG 1:** Responsible and socially sensitive leaders | **PLO 1.1** Demonstrate social sensitivity. | | | | |
| | **PLO1.2** Recommend responsible solutions to management problems. | | | | |
| | **PLO1.3** Demonstrate self-awareness and self-development. | | | | |
| **CG 2:** Capable of managing in a global environment | **PLO 2.1:** Interpret key aspects of the Indian and global business environment. | | | | |
| | **PLO 2.2** Analyse the Indian and global business environment for decision making. | | | | |
| **CG 3:** Proficient in functional and cross-functional business areas | **(PGDM) PLO 3.1** Demonstrate depth in selected functional areas of business. | ✓✓✓ | XXX | CLO 1 & 2 | Quiz 1–15%<br>Quiz 2–15%<br>GA – 40%<br>ET- 30% |
| | **(PGDM-BM)** PLO 3.1 Demonstrate breadth across functional areas of business. | | | | |
| | **PLO 3.2** Integrate across functional areas for decision making. | ✓✓ | XX | CLO 3 | GA – 40% |
| **CG 4:** Able to make reasoned decisions | **PLO 4.1** Examine business problems and alternative solutions. | ✓✓✓ | XXX | CLO 3 | GA – 40%<br>ET – 30% |
| | **PLO 4.2:** Recommend reasoned and actionable solutions. | ✓✓✓ | XXX | CLO 3 | GA – 40%<br>ET – 30% |
| | **PLO 4.3** Demonstrate clarity in thinking and articulation. | ✓✓ | XX | CLO 3 | GA – 40% |
| **CG 5:** Innovative leadership | **PLO 5.1** Understand principles and practices for innovation. | ✓✓ | XX | CLO 2 | GA – 40% |
| | **PLO 5.2** Develop innovative solutions in various managerial contexts. | ✓✓ | XX | CLO 3 | GA – 40% |

***Please only tick the most relevant CGs and PLOs**

Codes for AOL Table

| Codes for Teaching Learning Activities (TLA) | | | |
|---|---|---|---|
| Objective not covered | Objective barely covered (concept introduced) | Objective is addressed (concept practised), but needs further development | Very strong-TLAs designed to promote thorough, deep and active learning |
| - | ✓ | ✓✓ | ✓✓✓ |
| **Code for Assessment (Asmt.)** | | | |
| Objective not covered | Objective barely covered | Objective is addressed, but would need further development to demonstrate learning | Very strong - thorough, deep assessment of objective that would clearly demonstrate learning |
| - | X | XX | XXX |

**SDGs mapped to the course (Descriptions appended)**

| S.No. | UN SDG Goals | UN SGD Target | Please tick the applicable |
|---|---|---|---|
| 1 | Goal 4: Quality Education | 4.7 | ✓ |
| 2 | Goal 8: Decent Work and Economic Growth | 8.2, 8.3 | |
| 3 | Goal 9: Industry, Innovation & Infrastructure | 9.1, 9.4, 9. b | |
| 4 | Goal 10: Reduced Inequalities | 10.2, 10.3 | |
| 5 | Goal 12: Responsible Consumption and Production | 12.6 | |
| 6 | Goal 16: Peace, Justice, and Strong Institutions | 16.6, 16.7 | |
| 7 | Goal 17: Partnerships to achieve the Goals | 17.16, 17.17 | |

| Sessions/Cases/Discussion centered around ERS | | |
|---|---|---|
| **Content** | **Session No.** | **Any case/reading used** |
| **Ethics** | 8-9, 17-18 | 13 Principles for Using AI Responsibly. Harvard Business Review (2023) |
| **Responsibility** | 17-18 | 13 Principles for Using AI Responsibly. Harvard Business Review (2023) |
| **Sustainability** | | |

**Faculty Contact Preference**

**Office Hours -** Thursday 3pm-4pm

**Email –** aishvarya@spjimr.org

**Call/WhatsApp –** NA

**By Appointment -**